

Seek Wander

Curated Luxury Journeys

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This document defines the complete product requirements for **Seek Wander** — a luxury AI travel concierge. It covers product vision, user personas, feature specifications, business model, and success metrics. All features described herein are implemented and live in production unless explicitly marked **[PHASE: NEXT]** or **[PHASE: FUTURE]**.

Executive Summary [PHASE: COMPLETE — All 11 Phases Live]

Seek Wander transforms how discerning travellers plan their journeys. Users describe their preferences through an intelligent 7-field intake form; within 15–20 seconds, the system returns a fully-verified, editorial-quality multi-day itinerary — complete with GPS-accurate activities, Michelin-calibre restaurant recommendations, curated hotel stays, real transit estimates, and destination photography.

The product is built on a **zero-hallucination data model**: every location generated by the AI is immediately enriched and verified via **Google Places API** before being shown to the user. Ratings, opening hours, and photos are ground-truthed in real time. The word ‘AI’ does not appear anywhere in the user-facing product — the technology is the invisible engine; the luxury outcome is the headline.

Generation time	< 20 seconds	15–20s (p50)
Data accuracy	100% real places	Google Places verified
Cost per generation	< \$0.15	\$0.05–\$0.14 (60% cache)
Mobile experience	Full PWA	Installable, offline-capable
Affiliate integration	Booking.com live	AID 4013143 active
Uptime	99.9%	Vercel Edge (SLA guaranteed)

Product Vision & Positioning

Vision: The world’s most elegant travel planning experience — where every interaction feels like speaking with a seasoned concierge at a five-star hotel.

! POSITIONING RULE

The word 'AI' is intentionally absent from all user-facing copy, manifests, metadata, and UI text. The product is positioned around the luxury outcome ('Curated Luxury Journeys'), not the technology. **claude-sonnet-4-6** is the invisible engine — never the headline.

Brand Pillars

Editorial Aesthetic	Every screen feels like a luxury magazine spread. No border-radius. Paper tones (#F5F0E8), ink blacks (#0A0A0A), burnt-orange accents (#C2410C). Cormorant Garamond italic headings, DM Sans body copy.
Zero Hallucination	AI generates structure; Google Places API verifies every location. Users only see real, operational places with real photos and ratings.
Invisible Technology	The product outcome — a curated journey — is the brand. The underlying AI model is never referenced in user-facing copy.
Concierge Velocity	From preferences to complete itinerary in under 20 seconds. The speed of technology with the quality of a human expert.
Offline Resilience	Saved itineraries are accessible without connectivity via localStorage vault (seek_wander_archive). Installs as a PWA via @serwist/next .

Target Audience

The Affluent Explorer

Demographics	35–55, HHI \$200k+, frequent international traveller
Pain Points	Existing tools (TripAdvisor, Google Maps) are generic and overwhelming. ChatGPT produces geographically inaccurate results with no photo verification.
Core Need	A curated, verified, beautiful itinerary that respects their taste — fast, without requiring research expertise.

The Detail-Oriented Planner

Demographics	28–45, urban professional, values experience over possession
Pain Points	Spends hours researching before every trip. Anxious about missing hidden gems or making suboptimal restaurant choices.
Core Need	A single trusted source that surfaces real opening hours, reservation requirements, and realistic transit times.

The Luxury Couple / Honeymoon Planner

Demographics	28–40, planning once-in-a-lifetime trips, budget-agnostic
Pain Points	Every result feels touristy. Hard to find experiences that feel personal. PDF itineraries from travel agencies are expensive and impersonal.
Core Need	A shareable itinerary dossier they can export as a branded PDF and send to their partner.

Core Feature Specifications [STATUS: PRODUCTION READY]

4.1 — 7-Field Concierge Intake Form

Staged inline expansion — each stage unlocks only after the previous is completed, mirroring a concierge conversation. No page reloads. All validation is instant. Destination resolved via **Google Places Autocomplete** with GPS lat/lng capture.

1	Destination	Google Places Autocomplete	GPS lat/lng + placeId captured; Zod-validated on submit
2	Dates	Dual date picker	getLocalToday() for timezone safety; duration badge computed client-side
3	Travel Party	Pill selector	Solo / Couple / Family / Group
4	Pace	Card selector	Relaxed (3–4/day) / Moderate (4–5/day) / Packed (6–7/day)
5	Budget Tier	Card selector	Premium \$\$ / Luxury \$\$\$ / Ultra-Luxury \$\$\$\$
6	Dietary	Multi-select	7 options; Halal/Kosher/GF trigger extended dietaryNote in prompt
7	Accommodation	Toggle + optional field	needed → AI recommends stays; booked → user-provided hotel name

4.2 — AI Itinerary Generation Pipeline

Step 1: Zod Validation	ItinerarySchema.safeParse() — 12 fields validated before any computation. Returns 400 with per-field fieldErrors .
Step 2: Rate Limit	Upstash Redis slidingWindow(5, '1 h') per Clerk userId (auth) or IP (unauth). Returns 429 + X-RateLimit-* headers. Zero Anthropic API cost on rate-limited requests.
Step 3: AI Generation	claude-sonnet-4-6 with injection-resistant SYSTEM_PROMPT (passed as system: param, not in messages[]). Returns chronological timeline[] per day with real place names + GPS.
Step 4: JSON Repair	sanitizeJson() regex sweep (strips markdown fences + trailing commas). If JSON.parse() still fails → fallback JSON_REPAIR_PROMPT Anthropic call. Both failures return clean 500.
Step 5: Place Enrichment	enrichPlace() per timeline item — PlaceCache check first (30-day TTL). Miss → Google Places Text Search + Place Details . Adds: photoUrl, rating, openNow, hoursOpen, priceLevel.

Step 6: Transit	Pure Haversine formula — local math, zero API calls. Walking 5 km/h, driving 25 km/h between consecutive stops.
Step 7: Cost Logging	CostLog.create() awaited before <code>Response.json()</code> . Vercel freezes the function on response return — awaiting guarantees financial record integrity.

4.3 — Split-Screen Itinerary Display

Desktop: 55% scrollable editorial timeline (left) + 45% sticky **Google Map** (right). Mobile: full-width timeline with collapsible h-52 map banner. Tabbed day navigation with **AnimatePresence mode="wait"** transitions.

- **TimelineCard**: unified card for activities AND meals — photo, name, description, rating, open/closed badge, priceLevel, transit connector
- **StayCard**: Booking.com affiliate card for recommended hotels (AID 4013143)
- **Hidden Gem**: 1 per day — places 95% of tourists never find
- Day-centric SVG map markers with colour-coded legend (5 palette slots, keyed by day number not activity type)
- Polyline connecting all stops in chronological order: strokeColor #0A0A0A, strokeOpacity 0.08

4.4 — Trip Archive (/trips)

- Clerk-authenticated — only the owning user sees their trips
- IDOR-protected — `/trips/[id]` enforces `userId` ownership on every fetch; neutral `notFound()` for both missing and wrong-owner
- **useOfflineTrips** hook caches archive in **localStorage** (**seek_wander_archive**) for offline access
- Offline banner shown when `navigator.onLine` is false or `/api/trips` fetch fails
- **UnauthenticatedState** 'Velvet Rope' — editorial gate, Clerk modal sign-in (no redirect pages)
- **EmptyTripsState** 'Inspiration Hub' — 3 destination teasers instead of a blank screen

4.5 — PDF Export & Public Sharing

- **PDF Export**: `window.print()` — zero dependencies. Tailwind print: modifiers hide nav, map, tabs. All days rendered sequentially. Framer Motion opacity overridden in `@media print`.
- Branded dossier header in print: Seek Wander wordmark, destination, date — hidden on screen (hidden print:block).
- **Public Sharing**: `/shared/[id]` — intentionally public, no auth. Dynamic OG metadata (title, editorial description, destination photo) for iMessage/Slack/WhatsApp previews.
- **ShareButton**: `navigator.share()` (mobile native) with clipboard fallback + **Sonner** toast confirmation.

Business Model

Revenue Streams

Affiliate — Hotels	Booking.com AID 4013143 . Every recommended stay links to Booking.com search results. Commission on completed bookings via Booking.com affiliate programme. URL pattern: booking.com/searchresults.html?ss={query}&aid;=4013143	Live
Stripe Paywall (Phase 12)	\$4.99 per generation after free tier (1 free generation per account). Stripe Checkout — no credit card required to browse.	Planned
Brand Partnerships (Phase 13)	Destination tourism boards, luxury hotel chains paying for featured placement in recommended stays and editorial content.	Future

Booking.com Affiliate — Technical Integration

Affiliate ID	4013143 — registered to Seek Wander Booking.com partner account
Utility function	src/lib/affiliate.ts — createAffiliateUrl(hotelName, destination) returns encoded URL with &aid;=4013143 appended
URL pattern	https://www.booking.com/searchresults.html?ss={encodeURIComponent(hotelName + ' ' + destination)}&aid;=4013143
StayCard component	src/components/StayCard.tsx — motion.a, whileHover y:-2, ArrowUpRight icon CTA, micro-copy 'From [hotel name]', target='_blank' rel='noopener noreferrer'
Placement in ItineraryViewer	Rendered after the timeline section when itinerary.recommendedStays?.length > 0 . Each StayCard maps one recommendedStay to one affiliate link.
Print behaviour	print:hidden — StayCard section excluded from PDF dossier. Affiliate links have no value in a static PDF.
Attribution window	Booking.com: 30-day last-click cookie. Commission varies by property: typically 25–35% of Booking.com margin.

i NOTE — ACCOMMODATION BRANCHING

The AI prompt branches dynamically on **accommodationStatus**: when set to **'needed'**, the **SYSTEM_PROMPT** instructs **claude-sonnet-4-6** to generate a **recommendedStays[]** array (3 curated hotels with name, description, neighbourhood). When set to **'booked'**, the user-supplied **hotelName** is used as the itinerary base and no StayCard section renders. This prevents affiliate links appearing when the user has already committed.

Unit Economics

Anthropic claude-sonnet-4-6 input	\$3.00 / 1M tokens
Anthropic claude-sonnet-4-6 output	\$15.00 / 1M tokens
Google Places Text Search	\$0.032 / call
Google Place Details	\$0.017 / call
Typical generation (cache cold)	\$0.08 – \$0.14
Typical generation (60% cache hit)	\$0.05 – \$0.09
PlaceCache hit savings (per item)	\$0.049 (Text Search + Details)
Vercel Pro hosting	\$20 / month flat
Supabase Pro	\$25 / month flat
Upstash Redis	~\$0 (within free tier at current volume)

UX & Design Specifications

Design Tokens

paper	#F5F0E8	Primary backgrounds — warm off-white
paper-dark	#EDE8DC	Card backgrounds, subtle depth
ink	#0A0A0A	Primary text, borders
ink-light	#6B6B6B	Secondary text, captions, placeholders
emerald	#059669	Active states, OPEN badge
burnt-orange	#C2410C	CTA buttons, active tab indicator, CLOSED badge

Non-Negotiable Component Rules

- NO border-radius anywhere — rounded-none on every component (one explicit exception: MobileMenu hamburger uses rounded-full as a mobile touch affordance)
- Thin borders only — border border-black/5 or border-b border-ink/20
- **Framer Motion** on all sections — initial={{ opacity:0, y:24 }} animate={{ opacity:1, y:0 }} transition={{ duration:0.6, ease:'easeOut' }}
- Images: grayscale by default, full colour on hover (filter: grayscale(100%) → grayscale(0%), transition-all duration-700)
- Typography: **Cormorant Garamond** italic for headings, **DM Sans** for body/micro-copy
- Micro-copy: UPPERCASE, tracking-widest, DM Sans Bold
- HTML entities always: ' " … — never raw special characters in JSX

Notification System (Sonner)

All three generation-lifecycle toasts share **id: 'curate-task'** — Sonner mutates the same toast in place (no stacking). Styled to match brand: background #F5F0E8, borderRadius 0, fontFamily var(--font-dm-sans).

Loading	Consulting the concierge...	(spinner)	Fetch start
Success	Itinerary Prepared	Your bespoke journey is ready for review.	200 response
Error	Concierge Busy	Our desk is at capacity. Please try again in a moment.	Any error / 429
Save OK	Passport Updated	This journey has been saved to your archive.	Successful DB write
Save Err	Save Failed	Unable to save this journey. Please try again.	DB exception

Mobile & PWA Requirements [STATUS: INSTALLABLE PWA]

Service Worker Strategy (@serwist/next)

Source: `src/app/sw.ts` → compiled to `public/sw.js` by the `@serwist/next` webpack plugin. Disabled in `process.env.NODE_ENV === 'development'` to prevent stale cache contaminating hot-reload sessions.

CacheFirst	Google Places photo URLs (photoreference)	30-day TTL, 100-entry cap	Destination photos load instantly on repeat views; zero Google API calls after first fetch
defaultCache (Stale WhileRevalidate)	Next.js JS chunks, CSS bundles, fonts	Version-controlled	App shell loads offline; new deploys invalidate stale chunks
No cache	API routes (<code>/api/*</code>), Clerk auth endpoints	—	Dynamic data must always be fresh; auth tokens must not be cached

localStorage Vault — seek_wander_archive

The `useOfflineTrips` hook implements a two-phase offline strategy that enables the `/trips` archive to render without connectivity:

Cache key	<code>seek_wander_archive</code> — localStorage key on the user's device
Stored shape	<code>CachedTrip[]</code> — includes destination, days, createdAt, itineraryData (for editorial preview), and destination photoUrl (pre-resolved server-side via <code>MAPS_SERVER_KEY</code>)
Phase 1 — Immediate	On mount: check <code>navigator.onLine === false</code> . If offline, load from localStorage immediately. No network request attempted.
Phase 2 — Fetch-then-cache	If online: fetch <code>GET /api/trips</code> (Clerk-authed). On success: serialize response to localStorage. On fetch failure: fall through to localStorage, set <code>isOffline: true</code> .
UI indicator	<code>bg-ink text-paper</code> offline banner with <code>WifiOff</code> icon. Shown when <code>isOffline</code> is true regardless of <code>navigator.onLine</code> (covers server-down scenarios as well as true offline).
Cache invalidation	Every successful <code>/api/trips</code> response overwrites the cache. Stale data is shown only when no live response is available.

i NOTE — SERVER-SIDE PHOTO RESOLUTION

Photo URLs are computed server-side in **GET /api/trips** using **MAPS_SERVER_KEY**. The client (**TripsClient.tsx**) receives pre-resolved URL strings — it never holds or references the server API key. This is the canonical pattern for any server-only secret that produces client-visible URLs.

Responsive Breakpoints

- Mobile (< 768px): full-width single-column layout, map collapses to h-52 banner above timeline
- Desktop (≥ 768px): 55%/45% split-screen — timeline left, sticky map right
- Hamburger menu hidden at md+ breakpoint (desktop nav takes over)
- Print: single-column, all days sequential, map/nav/buttons hidden (print:hidden)

Security Requirements [STATUS: 10 CONTROLS ACTIVE]

! IMPORTANT — SECURITY PHILOSOPHY

All security controls are defence-in-depth: each layer assumes the previous has already been bypassed. **Zod** validates before any API call. **Upstash** rate-limits before any Anthropic call. **SYSTEM_PROMPT** injection defence operates independently of input validation. IDOR ownership checks are performed post-fetch, never via query scoping alone.

Zod ItinerarySchema validation on all POST bodies	Malformed payloads crashing AI pipeline or DB write	Zero 500 errors from bad user input; clean 400s with field-level feedback
Prompt Injection Defence (SYSTEM_PROMPT as system: param)	User-supplied destination/interests fields hijacking AI role or exfiltrating system state	Itinerary quality guaranteed regardless of adversarial input
Upstash Redis rate limiting: slidingWindow(5, '1 h')	API cost exhaustion attacks; a single actor flooding Anthropic spend	Hard cost ceiling per user; 429 returned before any AI token is consumed
AI JSON Self-Healing (sanitizeJson + repair prompt)	Malformed Claude output (markdown fences, trailing commas) reaching the UI as a broken render	0% broken itinerary displays; silent repair is invisible to users
Google API Key Isolation (client vs server key split)	Client key theft enabling unrestricted Google API usage billed to project	Client key restricted to HTTP referrers; MAPS_SERVER_KEY never transmitted to browser
IDOR Prevention (post-fetch userId ownership check)	User A accessing User B's saved trips via /trips/[id]	Data privacy compliance; neutral notFound() leaks no ownership info
Auth Guard on /api/trips (Clerk userId check)	Unauthenticated scraping of all user trip data	All queries scoped to { where: { userId } }; cross-user access structurally impossible
HTTP Security Headers (X-Frame-Options, nosniff, Referrer-Policy, Permissions-Policy)	Clickjacking, MIME-type confusion, cross-origin data leaks	Baseline security posture; passes OWASP header checklist
Admin Neutral 404 on /admin/metrics	Route enumeration revealing admin panel existence to non-admin users	notFound() (not redirect) means the route's existence is not disclosed

Cron Bearer Auth (Authorization: Bearer CRON_SECRET)	External actors triggering daily PlaceCache cleanup manually	Database integrity; Vercel injects CRON_SECRET on every scheduled run

Success Metrics & KPIs

Product Metrics

Generation completion rate	% of started generations that return a result	> 95%
Generation latency (p50)	Time from form submit to itinerary display	< 20s
Cache hit rate	PlaceCache hits / total enrichment calls	> 60% (steady state)
Cost per generation	AI + Google API spend per itinerary	< \$0.12
Save rate	% of generated itineraries saved to archive	> 30%
Share rate	% of saved trips shared via ShareButton	> 15%
PDF export rate	% of saved trips exported to PDF	> 10%
Mobile install rate	PWA installs / mobile unique visitors	> 5%
Affiliate CTR	Clicks on StayCard Booking.com links / itineraries with stays	> 20%

Business Metrics

Monthly affiliate revenue	Commission from Booking.com partner dashboard (AID 4013143)
Affiliate conversion rate	Confirmed bookings / affiliate clicks — target > 3%
Cost margin	(Affiliate rev + future Stripe) / Total AI + Google API cost
Monthly Active Users	Unique Clerk userIds with ≥ 1 generation in rolling 30 days

Phase Roadmap

1 — Core Engine	Next.js + Claude + Google Maps baseline	[PHASE: COMPLETE]
2 — Concierge UX	7-field intake, split-screen results, enrichment, Haversine transit	[PHASE: COMPLETE]
3 — Ultra-Luxury UI	Tabbed day nav, cost badges, transit connectors, day-centric markers	[PHASE: COMPLETE]
4 — Auth	Clerk v6, conditional ClerkProvider, NavbarAuth, custom 404	[PHASE: COMPLETE]
5 — Persistence	Prisma + Supabase, saveTrip, /trips, /trips/[id], IDOR enforcement	[PHASE: COMPLETE]
6 — PWA & Sharing	@serwist/next, /shared/[id] OG metadata, ShareButton, PDF export	[PHASE: COMPLETE]
7 — Timeline Refactor	timeline: TimelineItem[] canonical shape, normalizeDayPlan() shim	[PHASE: COMPLETE]
8 — Margin Protection	PlaceCache, Haversine, CostLog, /admin/metrics, Cron, Zod, headers	[PHASE: COMPLETE]
9 — Affiliate	Booking.com AID 4013143, StayCard, accommodation branching	[PHASE: COMPLETE]
10 — UX & Polish	GenerationLoader, Sonner toasts, UnauthenticatedState, EmptyTrips, MobileMenu portal fix	[PHASE: COMPLETE]
11 — Mobile & Resilience	useOfflineTrips, /api/trips, Upstash rate limiting, AI JSON self-healing, dynamic OG metadata	[PHASE: COMPLETE]
12 — Stripe Paywall	\$4.99 / generation after 1 free — Stripe Checkout, usage tracking	[PHASE: NEXT]
13 — Brand Partnerships	Featured placement for destination tourism boards + luxury hotel chains	[PHASE: FUTURE]